

CampusEats – External SOAP Partner Integration

CS543 Web Services – Assignment 3

Team Details

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1. Context

CampusEats integrates with an external payment gateway named **PaySecure Gateway** to process online card payments for food orders. The operation integrated from the external partner is **Charge**, which allows CampusEats to send an order payment request and receive either a successful transaction confirmation or a payment-related fault.

The rest of the CampusEats application uses REST services because its internal services are lightweight and resource-oriented. However, the payment partner uses SOAP because payment processing benefits from a strict service contract, structured XML messages, message-level credentials, and formal fault handling. The SOAP/WSDL contract clearly defines the exact request, response, fault structure, and HTTP binding required for communication between CampusEats and PaySecure.

External Partner: PaySecure Gateway

Integrated Operation: Charge

2. Partner WSDL

The PaySecure Gateway contract is defined in the file:

partner.wsdl

The WSDL contains all six required elements:

1. **Types** – Defines RequesterCredentials, ChargeRequest, ChargeResponse, and PaySecureFault.
2. **Messages** – Defines request, response, and fault messages.
3. **PortType** – Defines the Charge operation.
4. **Binding** – Specifies document-style SOAP over HTTP.
5. **Service** – Defines the PaySecureGatewayService.
6. **Port** – Provides the endpoint used by CampusEats.

The SOAP action defined for the Charge operation is:

<http://paysecure.example.com/actions/Charge>

The endpoint specified in the WSDL is:

<https://api.paysecure.example.com/gateway/v1/soap>

CampusEats uses this WSDL as the formal contract for constructing requests and interpreting responses and faults.

3. SOAP Message Exchange

The integration uses three SOAP messages, submitted as separate XML files.

3.1 SOAP Request

File:

soap-request.xml

CampusEats sends a SOAP request to the PaySecure Gateway. The credentials are included in the SOAP Header using the RequesterCredentials element.

The request body contains the ChargeRequest element with:

- Merchant Order ID
- Payment amount
- Currency code
- Card number
- Card expiry month
- Card expiry year
- Card holder name

Example order:

MerchantOrderId: CE-ORD-100245

Amount: 349.00 INR

The operation requested is Charge.

3.2 SOAP Success Response

File:

soap-response.xml

If the payment is processed successfully, PaySecure returns a ChargeResponse containing:

- Acknowledgement
- Transaction ID
- Merchant Order ID
- Amount charged
- Timestamp

Example response:

Acknowledgement: Success

TransactionId: PS-TXN-88213377

MerchantOrderId: CE-ORD-100245

AmountCharged: 349.00

CampusEats can use the returned TransactionId as the payment reference for the order.

3.3 SOAP Fault

File:

soap-fault.xml

If the payment cannot be completed, PaySecure returns a standard SOAP Fault. The structured details contain the PaySecureFault element.

The fault includes:

ErrorCode: card_declined

ErrorMessage: The card issuer declined this transaction.

MerchantOrderId: CE-ORD-100245

The SOAP fault allows CampusEats to distinguish between a successful payment response and a payment failure.

4. HTTP Binding

According to the WSDL, the Charge operation uses SOAP over HTTP.

CampusEats sends the SOAP request using an HTTP POST request to the endpoint defined by the service port.

POST /gateway/v1/soap HTTP/1.1

Host: api.paysecure.example.com

Content-Type: text/xml; charset=utf-8

SOAPAction: "<http://paysecure.example.com/actions/Charge>"

```
<tns:RequesterCredentials>
```

```
<tns:Username>campuseats_merchant</tns:Username>
<tns:Password>a1s2d3f4</tns:Password>
<tns:Signature>Xk29fLp0qRt8vZmN3jC7wYbH1eGdSa</tns:Signature>
</tns:RequesterCredentials>
</soapenv:Header>

<soapenv:Body>
<tns:ChargeRequest>
<tns:MerchantOrderId>CE-ORD-100245</tns:MerchantOrderId>
<tns:Amount>349.00</tns:Amount>
<tns:CurrencyCode>INR</tns:CurrencyCode>
<tns:CardNumber>4111111111111111</tns:CardNumber>
<tns:CardExpiryMonth>12</tns:CardExpiryMonth>
<tns:CardExpiryYear>2027</tns:CardExpiryYear>
<tns:CardHolderName>Suhan Patel</tns:CardHolderName>
</tns:ChargeRequest>
```

The complete endpoint URL is:

<https://api.paysecure.example.com/gateway/v1/soap>

The HTTP method is POST, and the SOAPAction value exactly matches the value defined in the WSDL binding:

<http://paysecure.example.com/actions/Charge>

The request body contains the SOAP Envelope defined in soap-request.xml.

5. Discovery

CampusEats obtains the PaySecure service contract through a modern service catalogue or registry.

During integration, the CampusEats development team searches the organisation's approved external-service catalogue for a payment processing service. The catalogue provides the service name, endpoint, and a pointer to the WSDL contract.

The WSDL can then be obtained from the partner's published contract location. CampusEats reads the WSDL and uses it to determine the supported operations, message formats, SOAP binding, SOAPAction, and service endpoint.

Registry / Catalogue Entry

Field	Value
Business	PaySecure Payment Services
Service	PaySecure Gateway Service
Operation	Charge
Endpoint	https://api.paysecure.example.com/gateway/v1/soap
Contract Type	SOAP/WSDL
WSDL / tModel-style Pointer	http://paysecure.example.com/wsd/ PaySecureGateway
Binding	SOAP over HTTP
SOAPAction	http://paysecure.example.com/actions/Charge

The registry entry acts as a modern discovery mechanism. Instead of operating a live UDDI server, CampusEats uses a catalogue record containing the information required to locate the service and obtain its WSDL contract.

6. Fault Mapping

The external PaySecure partner uses its own payment-specific fault vocabulary. CampusEats should not expose this partner-specific vocabulary directly through its own service contract.

For example, PaySecure may return:

ErrorCode: card_declined

CampusEats maps this external fault into its own domain-level error:

Partner Fault:
card_declined

↓

CampusEats Internal Error:
PAYMENT_DECLINED

↓

CampusEats placeOrder Response:

Payment declined. Please use another payment method.

The PaySecureFault and the value card_declined remain internal to the integration layer.

This mapping ensures that CampusEats maintains a stable and independent API contract. If the payment partner later changes its internal error vocabulary, only the integration layer needs to be updated. Students or clients using the CampusEats placeOrder operation continue to receive the CampusEats-defined error, such as:

payment declined

The external partner's terminology therefore does not leak into the CampusEats contract.

7. Integration Summary

The integration is contract-driven. The partner.wsdl defines the external partner interface, the SOAP request and response files demonstrate the message exchange, the HTTP binding shows how the request is transported, the registry entry describes service discovery, and the fault mapping protects the CampusEats contract from exposing external partner-specific errors.